Control the position of the Servo motor with the Arduino and the potentiometer.

Required Hardware:

1. Arduino Board
2. Servo Motor
3. 10k ohm Potentiometer
4. Jumper Wires
5. Mini Breadboard

Servo motors have three wires: power, ground, and signal. The power wire is typically red, and should be connected to the 5V pin on the Arduino board. The ground wire is typically black or brown and should be connected to a ground pin on the board. The signal pin is typically yellow or orange and should be connected to pin 9 on the board.

The potentiometer should be wired so that its two outer pins are connected to power (+5V) and ground, and its middle pin is connected to analog input 0 on the board.

You not connect directly the servo motor to the Arduino. I suggest you use external power to the servo. SG90 Mini Servo motors can be used.

Program:

```
#include <Servo.h>     // add servo library
Servo myservo;          // create servo object to control a servo
int potpin = 0;         // analog pin used to connect the potentiometer
int val;                // variable to read the value from the analog pin

void setup()
{
  myservo.attach(9);  // attaches the servo on pin 9 to the servo object
}

void loop()
{
  val = analogRead(potpin);          // reads the value of the potentiometer (value between 0 and 1023)
  val = map(val, 0, 1023, 0, 180); // scale it to use it with the servo (value between 0 and 180)
  myservo.write(val);                // sets the servo position according to the scaled value
  delay(500);                        // waits for the servo to get there
}
```